DM583

Data Mining

Introduction

Origins of Data Mining

Idea/Hypothesis → Experiments/Observations → Hypothesis Testing

- Historically, a **traditional setup** for *statistical data analysis* is such that one has an idea, or hypothesis, and they attempt to validate this *hypothesis* by testing it against observations, which are possibly the outcomes of carefully designed *experiments*

- Clearly, such traditional setup is still very important in many application scenarios

Adapted from:

DM868 DM870 DS804

(Arthur Zimek)

Knowledge Discovery from Data

Origins of Data Mining

Data Deluge

- Current Reality:

  Huge amounts of data collected in various forms and domains
- Manual analysis...?

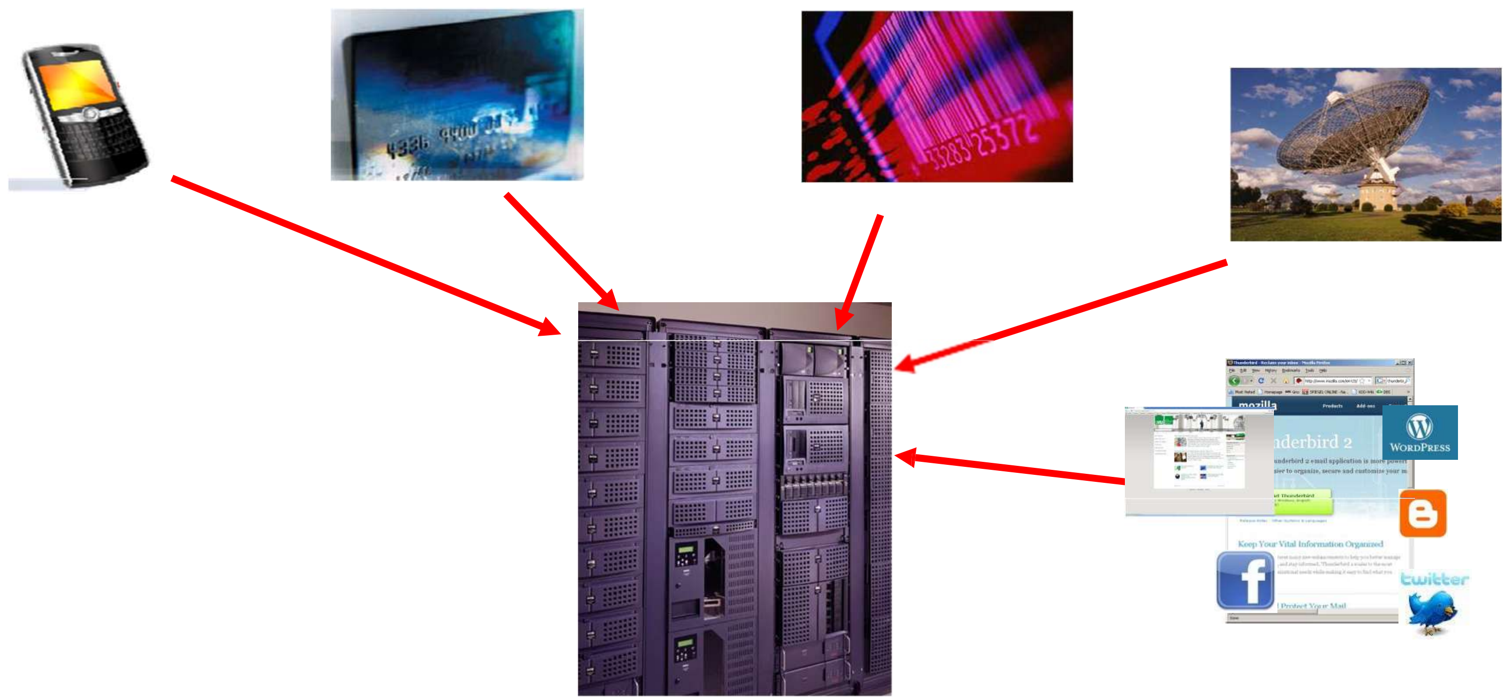
Adapted from:

DM868 DM870 DS804

(Arthur Zimek)

Knowledge Discovery from Data

Origins of Data Mining



- **Data deluge** requires algorithm-aided analysis

Data Analysis and Data Mining

Adapted from:

DM868 DM870 DS804

(Arthur Zimek)

Data Science

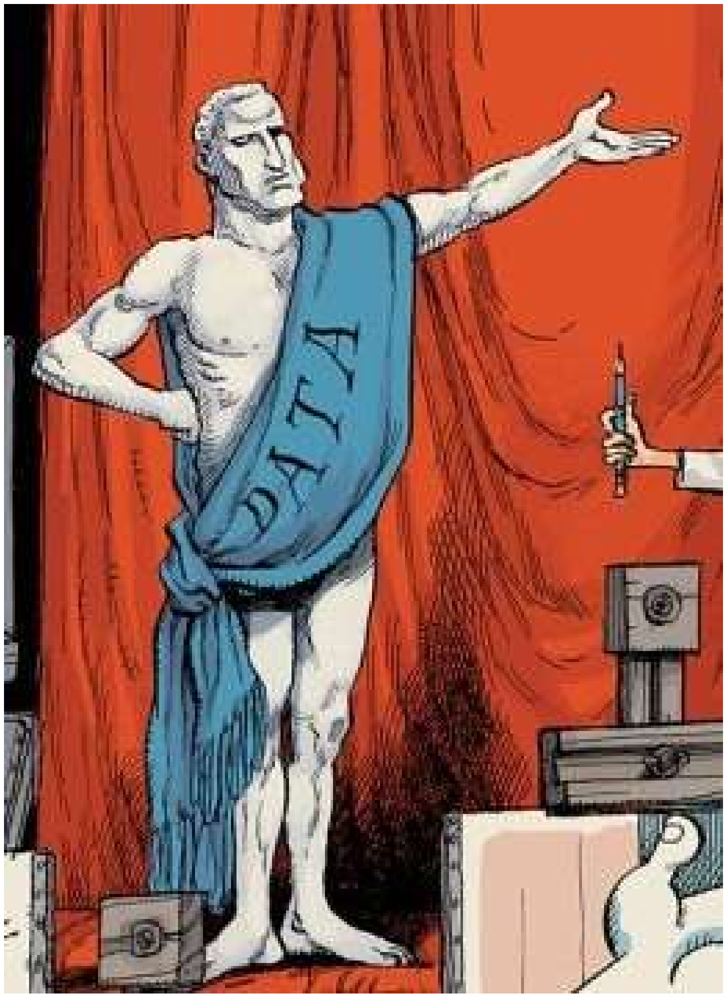


Illustration by David Parkins (detail).

Source: Silberzahn and Uhlmann [2015].

- Data Analysis: learning from data, finding patterns in data, understanding databases
- Data Mining: computational (algorithm-based) methods for data analysis

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Data Science

Data Analysis and Data Mining

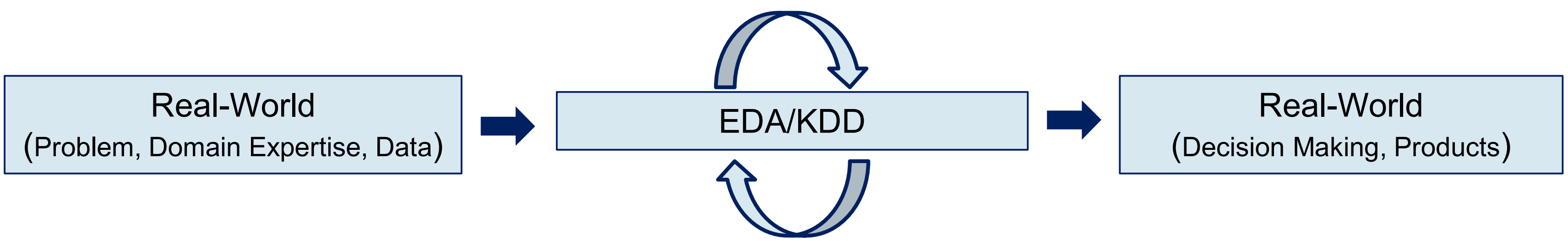
Illustration by David Parkins

Source: Silberzahn and Uhlmann [2015].

- Data Analysis: learning from data, finding patterns in data, understanding databases
- Data Mining: computational (algorithm-based) methods for data analysis
- Different methods may deliver different pictures of the data...

Data Mining, EDA and KDD

- Data mining is closely related to the concepts of:
  - **Exploratory Data Analysis** (**EDA**)
  - **Knowledge Discovery from Databases** (**KDD**); and
- The ultimate goal is to make sense of data



EDA (1970s)

- **Exploratory Data Analysis** (**EDA**):

  *"In statistics, **exploratory data analysis** (**EDA**) is an approach to analyzing data sets to summarize their main characteristics, often with visual methods. A statistical model can be used or not...*

  EDA was promoted by John Tukey to encourage statisticians to explore the data, and possibly formulate hypotheses that could lead to new data collection and experiments..."

Wikipedia Entry on **Exploratory Data Analysis**

KDD (1990s)

- **Knowledge Discovery from Databases (KDD):**
  - It is the process of identifying valid, novel, useful, and understandable patterns in data
  - It is a more recent concept related to EDA, yet broader in scope
    - Traditional EDA focuses mainly on visualization and descriptive statistics
    - KDD also encompass modern **data mining** and **machine learning** algorithms and models

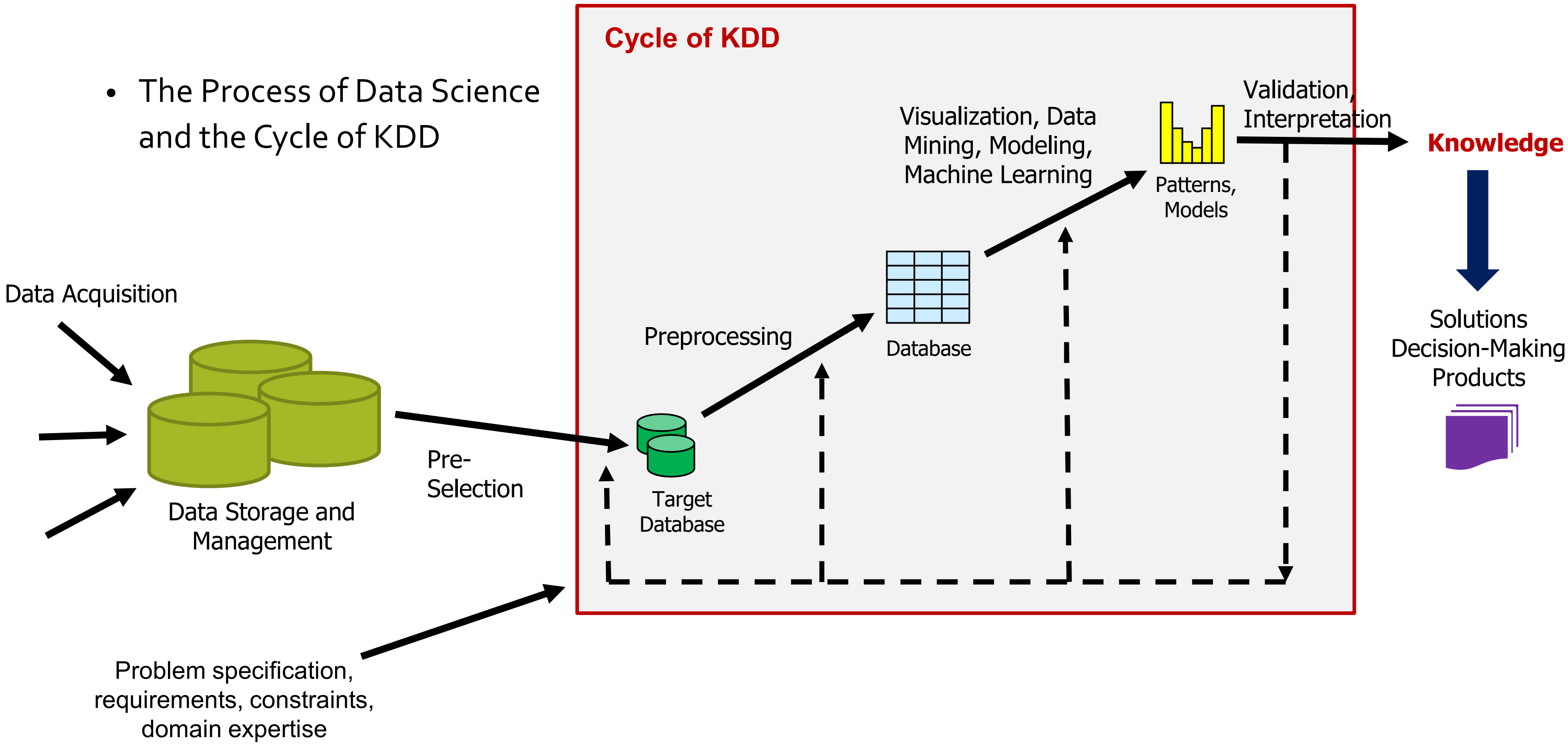
KDD (Original Definition)

"KDD is the nontrivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data." [Fayyad et al., 1996]

- *data*: set of facts (e.g., entries in a database)
- *pattern*: expression in some language to describe a data subset (e.g., mathematical model)
- *process*: can involve several steps or iterations
- *nontrivial*: more complex than search, inference, simple aggregations
- *valid*: applicable to new data with a certain degree of reliability
- *novel*: for the system / user
- *potentially useful*: beneficial for user of application
- *ultimately understandable*: if not immediately then given some post processing

KDD Cycle

- The Process of Data Science and the Cycle of KDD



Data Mining and Machine Learning

- **Machine Learning** (ML) highly overlaps with Data Mining (DM)
- Traditionally, ML is seen as a process whereby computers automatically learn how to solve tasks from previous experience
  - A computer is expected to learn from examples, improving its performance on the given task
- Unlike DM, ML is *not necessarily* intended for data analysis/KDD
  - E.g., self-driving cars, self-navigating robots, and other AI tasks
- Certain Tasks are both ML and DM tasks (e.g. regression, classification):
  - ML perspective generally focuses on prediction power/effectiveness only
  - DM perspective in these tasks can be different or broader…

Data Mining/ML: Training Strategies

- **Supervised Learning** (e.g., classification, regression, anomaly detection):
  - A dependent or output variable, Y, is modelled as a function of a collection of independent or input variable(s), X, so-called predictor(s)
  - For instance, credit risk as a function of customers' banking history and/or personal details
  - Past observations from both (X, Y) serve as a "teacher" for a model to be trained from data, and the training of such a model is referred to as supervised learning

Supervised Learning Task (Regression)

- **Regression**:
  - The dependent variable Y is real-valued (numeric, continuous)
  - It is presumed that it can be described as a function f of the predictor(s) X, i.e.

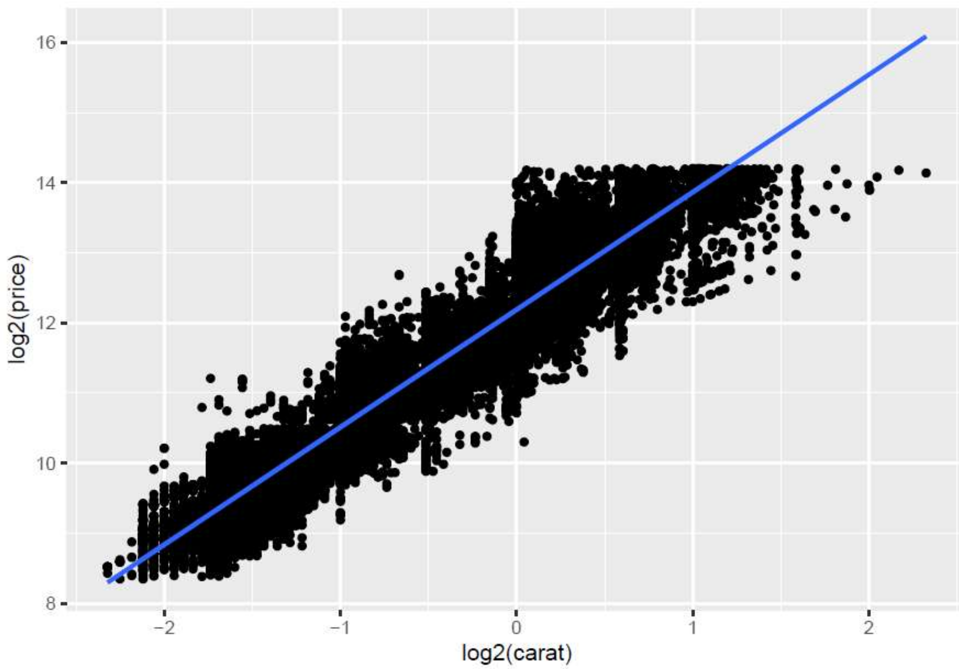
$$Y = f(X) + \varepsilon$$

    Except for a component ε that depends on unobserved variables (obs. error, noise)

  - Main goal is to learn a model $\hat{f}$ for the unknown (possibly non-linear) mapping f from a data set containing a representative collection of observations (X, Y)
  - **ML Perspective**: typically, only or mostly focused on *prediction* (e.g. *black-box* models)
  - **DM (*Potentially* Extended) Perspective**: e.g., which predictors most influence the output? how do such predictors influence the output? what do $\hat{f}$ tell us about the domain?

Supervised Learning Task (Regression)

- **(Simple Linear) Regression Example:** $Y = f(X) + \varepsilon \;\rightarrow\; \hat{Y} = \hat{f}(X) = b_0 + b_1 X_1$



Regression Example: `diamonds` dataset

Supervised Learning Task (Classification)

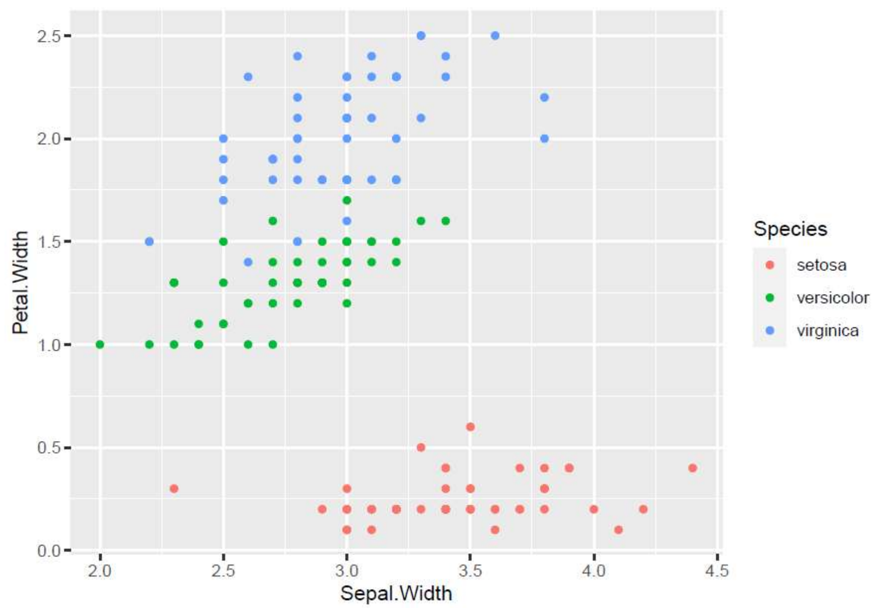
- **Classification**:
  - The dependent variable *Y* is categorical
    - It takes on a finite set of nominal values (*class labels*)
  - It is also presumed that it can be described as a function *f* of the predictor(s) *X*, except for a component ε that depends on unobserved variables (observation error, noise)
  - Main goal is to learn a model $\hat{f}$ for the unknown mapping *f* from a data set containing a representative collection of observations (*X*, *Y*)
  - **ML Perspective**: typically, only or mostly focused on *prediction* (e.g. *black-box* models)
  - **DM (*Potentially* Extended) Perspective**: e.g., which predictors most influence the output? how do such predictors influence the output? what do $\hat{f}$ tell us about the domain?

Supervised Learning Task (Classification)

- **Classification Example:** $Y = f(X) + \varepsilon \;\rightarrow\; \hat{Y} = \hat{f}(X)$

```r
head(iris, 10)
```

```
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1           5.1         3.5          1.4         0.2  setosa
## 2           4.9         3.0          1.4         0.2  setosa
## 3           4.7         3.2          1.3         0.2  setosa
## 4           4.6         3.1          1.5         0.2  setosa
## 5           5.0         3.6          1.4         0.2  setosa
## 6           5.4         3.9          1.7         0.4  setosa
## 7           4.6         3.4          1.4         0.3  setosa
## 8           5.0         3.4          1.5         0.2  setosa
## 9           4.4         2.9          1.4         0.2  setosa
## 10          4.9         3.1          1.5         0.1  setosa
```



Classification Example: `iris` dataset

Data Mining/ML: Training Strategies

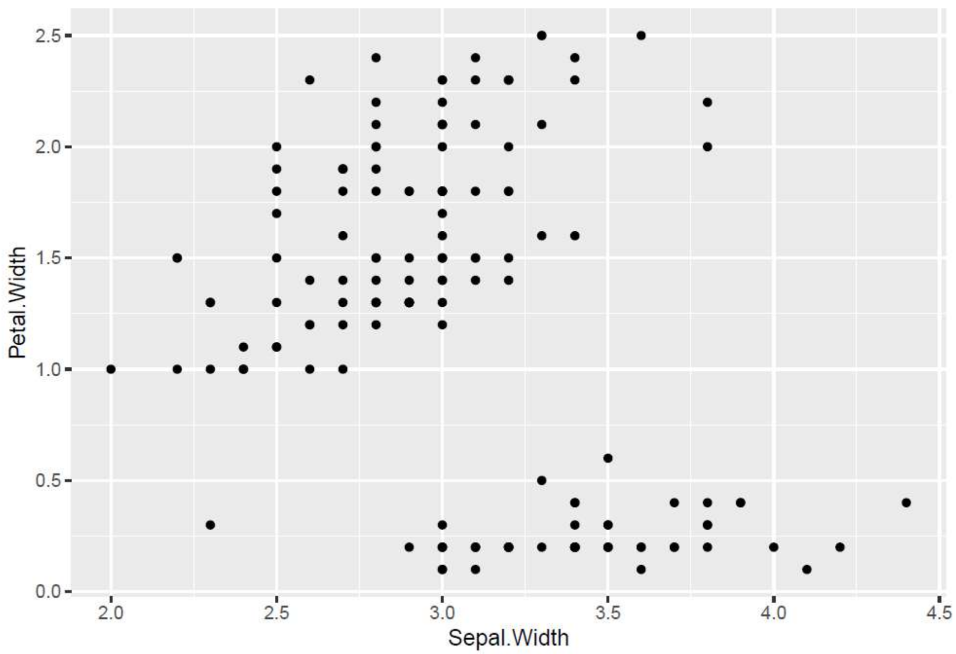
- **Unsupervised Learning** (e.g., clustering, outlier detection, association rules):
  - There is no dependent/output variable, only the input variables X
  - Patterns are learnt from X only
    - without explicit teacher/guidance on desired outputs
  - e.g., groups of similar customers based on their purchase behavior and personal details

Unsupervised Learning Task (Clustering)

- **Clustering**
  - In clustering we are interested in finding groups (clusters) of observations that are somehow more "similar" or "related" to each other than observations in other groups
  - Cluster analysis is a well-established field of statistics at least since the 1940/50s
    - e.g. market segmentation problem
  - It is sometimes referred to as **unsupervised classification**

Unsupervised Learning Task (Clustering)

- **Clustering Example:** 2 major clusters



Clustering Example: `iris` dataset

```
head(iris, 10)
```

```
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1           5.1         3.5          1.4         0.2  setosa
## 2           4.9         3.0          1.4         0.2  setosa
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## 8           5.0         3.4          1.5         0.2  setosa
## 9           4.4         2.9          1.4         0.2  setosa
## 10          4.9         3.1          1.5         0.1  setosa
```

Unsupervised Learning Task (Outliers)

- **Outlier Detection Example:**

```r
head(iris, 10)
```

```
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1           5.1         3.5          1.4         0.2  setosa
## 2           4.9         3.0          1.4         0.2  setosa
## 3           4.7         3.2          1.3         0.2  setosa
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## 9           4.4         2.9          1.4         0.2  setosa
## 10          4.9         3.1          1.5         0.1  setosa
```

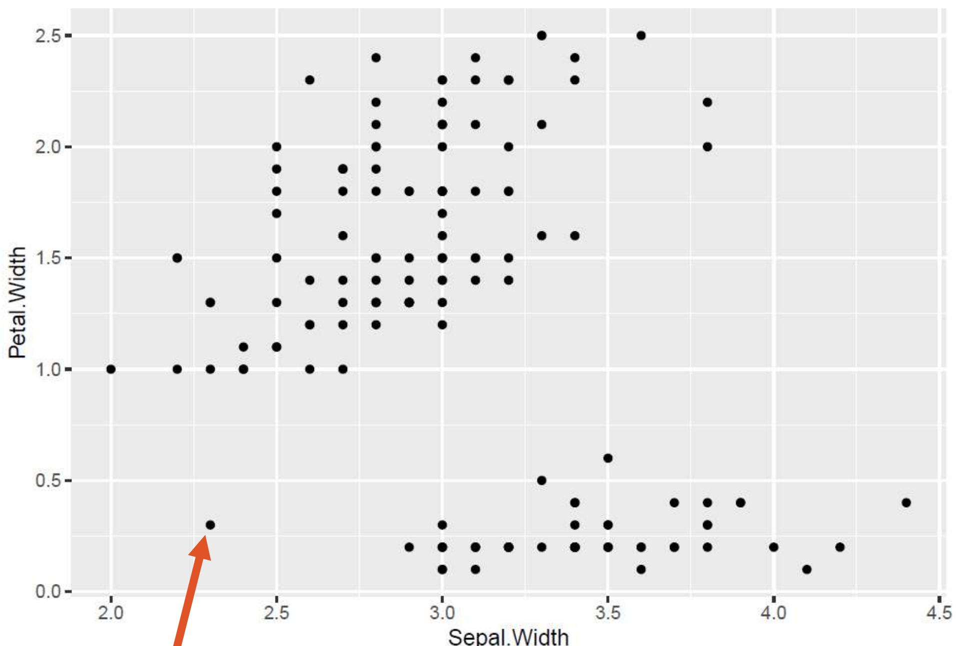

Clustering Example: `iris` dataset

Abnormality (error, fraud, …) ??

Unsupervised Learning Task
(Association Rules)

a,**b,c,d**,e
b,c,d
a,**b,c,d**
c,d,f
a,**b,c,d**,e
a,c,d
a,c,e,f
c,d,e,f
a,**b,c,d**,f
a,b,e,f

In 5 out of 10 (50%) cases,

b,c,d

occur together.

In 5 cases we have **b,c**, and in all those 5 cases we also have **d**:

Rule with 100% confidence:

If **b,c** *are in the set, then also* **d** *is in the set.*

Sample of Other, Specialized DM Tasks

- Recommendation
- Community Detection in Social Networks
- Sentiment Analysis
- Link Analysis/Prediction
- …

References

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2. J. Han, M. Kamber, and J. Pei. "*Data Mining: Concepts and Techniques*", Morgan Kaufmann, 3rd edition, 2011
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4. M. J. Zaki and W. Meira JR. "*Data Mining and Analysis: Fundamental Concepts and Algorithms*", Cambridge Univ. Press, 1st Edition, 2014